
EXTENDED ABSTRACT

PROJECT TITLE

MODULE NAME
Lectoraat Data Intelligence
Zuyd Hogeschool

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Abstract

Maximum of 200 words: The abstract is a summary of the extended abstract as a whole. This implies that at least the following components from the extended abstract are included in summary form in the abstract: reason, objective, method, results and conclusion

1 Introduction

Maximum of 800 words (for background and goal combined)

1.1 Background

The background describes the context in which the assignment was carried out:

- The companies/organizations involved and the products and/or services they provide
- The stakeholders, including the client
- The problem, need or opportunity that is the reason for the assignment

1.2 Goal

The goal logically follows from the background and describes the assignment to be carried out in a concise but concrete manner:

- The goal to be achieved (for the client)
- Any further specification of that goal in the form of boundary conditions and/or requirements (if present)
- The specific artefact(s) to be delivered, if already known at the start of the work

2 Theoretical Framework

Maximum of 800 words. Not required for courses other than the Design Science Research minor.

The theoretical framework describes the results of a (quick) literature scan with which the assignment to be carried out is classified on the basis of:

- The problem, the opportunity or the need in the application domain,
- The (possible) solution(s)
- The innovative character

The results of the literature scan show on the one hand to which category the problem belongs and on the other hand which (from existing work, i.e. broadly supported) uncertainties (with regard to the approach and/or end result) exist within the assignment.

3 Methods

Maximum of 1200 words.

The method describes the entire approach, so that others can in principle reproduce the work. The description of the method takes place at 3 levels.

First, the chosen overall method is described, in line with the objective of the assignment. Within the HBO-ICT program for example, assignments often consist of creating a (partly) innovative artifact (product or service). Design Science Research (DSR) is the standard method within the program for such design-oriented research.

Subsequently, the concrete steps carried out within the chosen method are described in the form of phases and/or activities. It becomes clear whether these were carried out sequentially, parallel and/or iteratively. An illustration is highly recommended here. Specific from the DSR method should in any case be included: identifying requirements, carrying out grounding, designing, developing and testing the product or service and disseminating the results of the work carried out. The field test from the DSR framework is optional, as a rule students will not arrive at a product that is ripe enough for a field test within the boundaries of the education.

Last but not least, the methods/methods used specifically within the phase(s) / activity(ies) are described concretely and in depth, where possible consisting of best or good practices. It is always important that the choices made are substantiated in relation to the context and objective of the assignment

4 Results

Maximum of 1200 words.

The results describe in depth the actual, substantive results achieved, logically following from the method used. Name the concrete artefact that (for the client) forms the end product of the delivered work. Indicate what type of artefact it is, for example a design, an architecture, a model or a software application. Then describe in depth the content of the artefact based on the (which?) requirements, the (which?) relevant insights from grounding and the (which?) relevant insights from activities within the design cycle. Use illustrations where useful, this is very clarifying compared to texts in (long) paragraphs only. Where possible, also use lists and/or tables for readability.

4.1 Examples

Inserting an image can be done as followS:

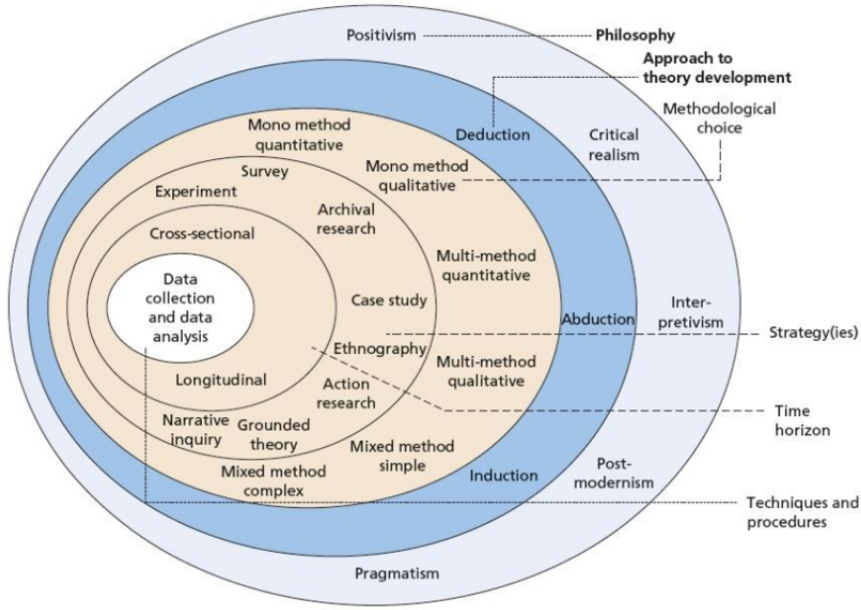


Figure 1: The research onion (adopted from [1]). [for illustration purposes only, so remove from the EA].

You may want to insert Python code snippets to show the design of your algorithm. We created a Python macro in the preamble that makes it easy to insert snippets of code. This environment takes two arguments `\begin{python}(caption)(label)` where:

1. *caption* can be used to provide a description for this code listing
2. *label* can be used as an identifier to reference this listing in the text

As an example, consider listing 1.

```
1 def foo(x: int, y: float) -> str:
2     z = x + y
3     return "Hello " + z
```

Code 1: Important algorithm

There is also the option to write pseudo code in L^AT_EX. See the following example in algorithm 1.

Algorithm 1 Simplified version of the Wave Function Collapse algorithm

Input: Sample image *img*;
Output: Rendered output image

```
1: function WFC(img)
2:   patterns := ExtractFromSample(img);
3:   ConstructIndex(patterns);
4:   InitialiseGrid(patterns);
5:   repeat
6:     p := Observe();
7:     if p = null then
8:       break;
9:     Collapse(p);
10:    Propagate();
11:  until all variables are observed
12:  return ObservationsToOutput();
```

Often, you evaluate your artifacts quantitatively, elicit requirements from stakeholders, or may aggregate guidelines from the contemporary literature. In these cases, it can be useful to display these results in a table. See table 1, 2, and 3 for various table examples.

Tabel 1: Confusion matrix for two classes.

	<i>Predicted class</i>		
<i>True class</i>	Positive	Negative	<i>Total</i>
Positive	<i>tp</i> : true positive	<i>fn</i> : false negative	<i>p</i>
Negative	<i>fp</i> : false positive	<i>tn</i> : true negative	<i>n</i>
<i>Total</i>	<i>p'</i>	<i>n'</i>	<i>N</i>

Tabel 2: Evaluation results of the selected hypothesis classes over 98 repetitions of training/testing splits for both the baseline as well as the difference datasets. Mean AUC and cross-entropy loss are reported.

Model	Baseline		Difference	
	<i>AUC</i>	<i>Cross-entropy</i>	<i>AUC</i>	<i>Cross-entropy</i>
Random Forest	0.605 ± 0.010	12.227 ± 0.332	0.609 ± 0.012	12.281 ± 0.377
XGBoost Classifier	0.617 ± 0.014	12.474 ± 0.456	0.612 ± 0.012	12.314 ± 0.394
Support Vector Machine	0.560 ± 0.017	15.220 ± 0.631	0.511 ± 0.007	14.453 ± 0.242
Multi-Layer Perceptron	0.622 ± 0.012	12.096 ± 0.423	0.618 ± 0.011	12.155 ± 0.355

Tabel 3: Hyperparameter search space for the Random Forest classifier.

Parameter	Description	Values
<code>n_estimators</code>	Number of decision tree estimators.	[50, 5000] with steps of 50.
<code>min_samples_leaf</code>	Minimum number of instances required at a leaf node.	[5, 50] with steps of 5.
<code>max_depth</code>	Maximum depth of the decision trees.	[4, 24]
<code>criterion</code>	Function to measure quality of a split.	'gini' or 'entropy'
<code>max_features</code>	For d features, it is the $k < d$ -features to consider when looking for the best split.	$\sqrt{d}$ or $\log_2(d)$, or fractions of d : 0.1, 0.3, 0.5, 0.7, 0.9.
<code>bootstrap</code>	Whether bootstrap samples are used when building trees.	True or False

5 Discussion

Maximum of 400 words.

The discussion describes a critical view (reflection) of the author(s) on the work delivered as a whole:

- Which aspects have contributed positively to the results achieved and why?
- Which aspects have been limiting for the results achieved and why?

Aspects can be considered broadly here. For example, process choices (method), design decisions (results) or other (whether or not externally determined) aspects that have influenced the results

6 Conclusions

Maximum of 400 words. The conclusion describes the extent to which the original goal of the assignment has been met with the help of the achieved (factual, substantive) results. In addition, any recommendations for follow-up work are described:

- Which artefacts have/have not been realised?
- To what extent have any boundary conditions/requirements been met?
- To what extent can the results be used/used for the original purpose?
- What recommendations are made on the basis of this, both towards practice and any necessary follow-up research into previously carried out work?

Verwijzingen

- [1] S. Mardiana, “Modifying Research Onion for Information Systems Research,” *Solid State Technology*, jrg. 63, p. 1202–1210, dec 2020.

Bijlagen

A Additional material